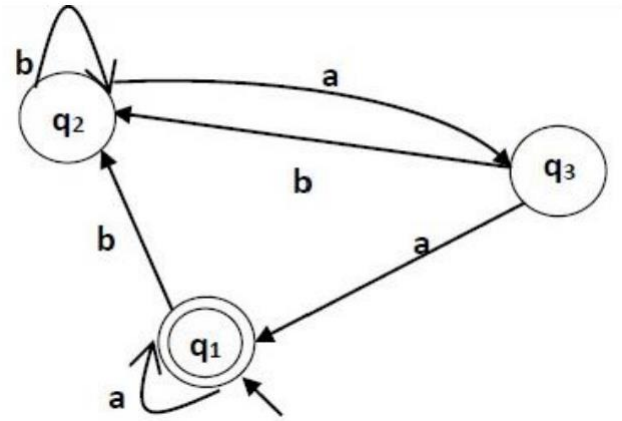
Roll Number: _____	
Thapar Institute of Engineering and Technology, Patiala	
Department of Computer Science and Engineering **END-Semester Examination**	
BE COE & CSE (Third Year)	**UCS701: Theory of Computation**
18th May 2024, 2:00 PM	**Time: 3Hours, Max Marks:40**
Instructors: Sunita Garhwal, Nidhi Kalra, Shashank Sheshar, Aditi Sharma, Rohit Ahuja.	

Note: Attempt any 5 questions. Attempt all questions (subparts) in sequence at one place. Assume missing data, if, any, suitably.

Q No.	Questions	Marks	CO	BL
Q1.	a) Given the Context-free grammar G: $S \rightarrow AB$ $A \rightarrow AA \mid AB \mid a$ $B \rightarrow CC$ $C \rightarrow b$ Apply CYK algorithm to determine whether the string $w = aaabb \in L(G)$ or not?	5	CO2	L3
	b) Construct deterministic finite automata for the language: $L = \{a^n \mid n \neq 2, n \neq 6, \text{over } \Sigma = \{a\}\}$.	3	CO1	L3
Q2.	a) Draw the flowchart for the language L={w a^{2m} b^{3m} w^R | w∈ {a,b}* and w^R is the reverse of w , m>0} accepted by the push-down machine. Construct the transition diagram for the above designed flowchart.	5 3	CO2 CO2	L4 L4
	b) Construct a Context-free grammar over ∑={a,b} for the language: L={$a^n b^m$ | 0<=2n<=m<=3n}.			
Q3.	a) Consider the context-free grammar G: S→ aAa| bBb | ^ A→ C | a B→ C | b C→ CDE | ^ D→ A | B | ab Convert the above grammar into Greibach Normal Form(GNF).	5	CO2	L3 L4
	b) Prove that L= {a^P | P is prime} is not a context-free language using Pumping Lemma .	3	CO3	
Q4.	a) Write down the logic for the design of a Turing machine to find the successor of a positive integer over ∑={0,1}. Design the Turing Machine for the above designed logic.(Input and Output should be in binary format).	4	CO3	L3
	b) Write down the logic for the design of a Turing machine for the language $L = \{a^n b^{2n} \mid n > 0\}$. Design the Turing Machine for the above designed logic.	4	CO3	L4
Q5.	a) Design a Post Machine for a language L={a^{2n} b^{3n} c^{2n} | n>=0}.	4	CO4	L3
	b) Check for the string w=*aabbc* , whether the following	4	CO4	L4

[P. T. O.]

	grammar is ambiguous or not: S→ABC | ab A→aA | B | ^ B→Sb |b| ^ C→cC |bS |^			
Q6.	a) Design a DFA for L={w|w∈{1,0,2}*, where fourth and third last symbol in w are 2 and 0 respectively}. b) Write down regular expression corresponding to the following finite automaton using Arden's Theorem.  	4 4	CO1 CO1	L4 L4

